

DINOSAUR TIME MACHINE

A Science Story About Prehistoric Life

Ages 6-12

Written by Daniel Tesfamariam

Science Story Series - Book 7

Travel back 230 million years and meet
the incredible creatures that ruled Earth!

Educational Content | Real Science | Amazing Stories

DINOSAUR TIME MACHINE

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For ages 6-12. Scientific facts based on current paleontological research.

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Chapter 1: The Time Machine

Twins Leo and Luna were exploring their grandpa's dusty attic one rainy afternoon. Behind old boxes and cobwebs, they found something incredible - a strange machine! It had dials, buttons, and a screen that said 'DESTINATION: 230,000,000 YEARS AGO.' 'It is a time machine!' Luna gasped. 'Should we try it?' Leo grinned. 'Absolutely! Let us visit the age of the dinosaurs!' They climbed in, pressed the big green button, and WHOOOOSH - they were gone!

THE SCIENCE: HOW WE KNOW ABOUT DINOSAURS

Fossils are preserved remains or traces of ancient life (bones, teeth, footprints). Scientists called paleontologists dig up and study fossils to learn about the past. Fossils form when an animal is buried quickly in sediment after death. Over millions of years, minerals replace the bone, turning it to stone. We can determine fossil ages using radioactive dating of the surrounding rock. Dinosaurs lived during the Mesozoic Era: Triassic, Jurassic, and Cretaceous periods.

SCIENCE FACT BOX

- * The word 'dinosaur' means 'terrible lizard' in Greek (coined in 1842).
- * Over 1,000 different dinosaur species have been named so far!
- * New dinosaur species are discovered about every 2 weeks somewhere in the world.

TRY THIS!

Make a fossil: press a shell or leaf into clay, remove it, fill the impression with plaster!

THINK ABOUT IT:

Why do we only find fossils of some animals and not all of them?

Chapter 2: The Triassic Period

The time machine landed with a THUD. Leo and Luna stepped out into a hot, dry world.
'Welcome to the Triassic Period - 230 million years ago!' the machine announced.
The landscape looked different: all the continents were joined in one supercontinent!
Small dinosaurs, only about the size of dogs, scurried through the ferns.
'These are the FIRST dinosaurs!' Luna pointed. 'They are so small!'
'But give them 165 million years,' Leo smiled, 'and they will rule the whole planet!'

THE SCIENCE: THE TRIASSIC WORLD

The Triassic Period lasted from 252 to 201 million years ago.
All continents were joined in one supercontinent called Pangaea.
The first dinosaurs appeared about 230 million years ago - they were small!
Early dinosaurs like Eoraptor were only about 3 feet long and walked on two legs.
Other animals lived alongside them: early mammals (tiny!), crocodile ancestors, and more.
The climate was mostly hot and dry with no polar ice caps.

SCIENCE FACT BOX

- * Pangaea means 'all lands' in Greek - it was one giant landmass!
- * The first dinosaurs were not the biggest - they had to compete with other reptiles.
- * Dinosaurs became dominant after a mass extinction wiped out their competitors.

TRY THIS!

Cut out the continents from a map and try to fit them together like a puzzle - that is Pangaea!

THINK ABOUT IT:

If all continents were once joined, how did they move apart? What force moved them?

Chapter 3: Giants of the Jurassic

The twins jumped forward in time to the Jurassic Period. The world had changed!
Suddenly, an ENORMOUS shadow fell over them. They looked up... and up... and UP!
A Brachiosaurus towered above the trees - taller than a five-story building!
'It is as long as three school buses!' Leo whispered, amazed.
The gentle giant munched on treetop leaves, unbothered by the tiny humans below.
A herd of them moved slowly across the lush green landscape like living mountains.

THE SCIENCE: GIANT HERBIVORES

Sauropods were the largest land animals EVER - some over 100 feet long!
They were herbivores (plant-eaters) with long necks to reach high treetops.
Brachiosaurus: 85 ft long, 40 ft tall, weighed 80 tons (like 11 elephants!).
Diplodocus: 90 ft long with a whip-like tail for defense against predators.
They swallowed stones (gastroliths) to help grind food in their stomachs!
Their long necks let them eat from trees that other dinosaurs could not reach.

SCIENCE FACT BOX

- * Sauropod footprints have been found that are over 5 feet across!
- * The biggest sauropod (Argentinosaurus) may have weighed 100 tons!
- * Some sauropods may have lived 100+ years based on bone growth ring analysis.

TRY THIS!

Measure 85 feet outside (about 28 big steps). That is how long a Brachiosaurus was!

THINK ABOUT IT:

How much food would a 100-foot-long animal need to eat every day?

Chapter 4: The Fierce Hunters

The twins traveled to the Cretaceous Period - 68 million years ago.
THUMP. THUMP. THUMP. The ground shook with heavy footsteps.
Through the trees appeared the most famous dinosaur of all: Tyrannosaurus Rex!
It was 40 feet long with teeth the size of bananas and jaws that could crush bone!
'RUN!' Leo yelled. But the T. Rex was not interested in them - it was chasing prey!
They watched from behind a fallen log as the mighty hunter sprinted past at 20 mph.

THE SCIENCE: CARNIVOROUS DINOSAURS

T. Rex: 40 ft long, 12 ft tall at the hip, 9 tons - the apex predator!
Its bite force was 12,800 pounds - strong enough to crush a car!
T. Rex had excellent eyesight and could see better than a hawk.
Velociraptor: smaller (6 ft) but hunted in packs and was incredibly smart.
Spinosaurus: possibly the largest carnivorous dinosaur (50+ ft), ate fish!
Carnivores had sharp, curved teeth for tearing meat (unlike flat herbivore teeth).

SCIENCE FACT BOX

- * T. Rex arms were only about 3 feet long - but could still lift 400 pounds each!
- * T. Rex could eat up to 500 pounds of meat in a single bite.
- * Real Velociraptors were only about 6 feet long and had feathers (not like in movies!).

TRY THIS!

Compare your teeth in a mirror: flat ones for grinding (herbivore) and pointed ones for tearing (carnivore)!

THINK ABOUT IT:

Was T. Rex a hunter, a scavenger, or both? What evidence would tell us?

Chapter 5: Flying Reptiles

Luna looked up and saw something incredible soaring through the Cretaceous sky.

A Pteranodon glided overhead with wings wider than a small airplane!

'Wait - are those flying dinosaurs?' Leo asked.

'Actually no!' Luna corrected. 'Pterosaurs were flying reptiles, NOT dinosaurs!'

'They lived alongside dinosaurs but belonged to a completely different group.'

The enormous creature swooped down toward the ocean, snatching fish from the waves.

THE SCIENCE: PTEROSAURS

Pterosaurs were flying reptiles that lived alongside dinosaurs - but were NOT dinosaurs!

Their wings were made of skin membrane stretched from an elongated finger to their body.

Quetzalcoatlus had a 36-foot wingspan - as wide as a fighter jet!

Despite their size, many pterosaurs were surprisingly lightweight (hollow bones).

Some pterosaurs were as small as sparrows; others were the largest flying animals ever.

They went extinct along with the dinosaurs 66 million years ago.

SCIENCE FACT BOX

- * Quetzalcoatlus stood as tall as a giraffe when on the ground!
- * Pterosaurs had excellent eyesight for hunting fish from high in the air.
- * The first pterosaur fossil was discovered in 1784 in Germany.

TRY THIS!

Make a paper airplane. How is its 'wing' different from a bird's wing or a pterosaur's wing?

THINK ABOUT IT:

What is the difference between pterosaurs, birds, and bats? They all fly differently!

Chapter 6: Ocean Giants

The time machine could also work underwater! The twins dove into the ancient sea.

Massive creatures swam through the dark water all around them.

A Mosasaurus glided past - 50 feet of powerful jaws and streamlined body!

'These are marine reptiles,' Luna explained. 'Also NOT technically dinosaurs.'

Long-necked Plesiosaurs paddled gracefully with their four flippers.

'The oceans were just as dangerous as the land!' Leo gulped.

THE SCIENCE: MARINE REPTILES

Marine reptiles ruled the oceans while dinosaurs ruled the land.

Mosasaurus: 50 ft long, top predator of Cretaceous seas, ate sharks and fish.

Plesiosaur: long neck, four flippers, some were 40+ feet long.

Ichthyosaur: shaped like dolphins, could swim up to 22 mph, gave live birth!

These ocean giants were air-breathing reptiles - they had to surface to breathe.

None of these marine reptiles were actually dinosaurs (dinosaurs lived only on land).

SCIENCE FACT BOX

- * Some Plesiosaurs had necks 23 feet long - half their total body length!
- * Loch Ness Monster legends may have been inspired by Plesiosaur fossils.
- * Mosasaurus is related to modern Komodo dragons and monitor lizards.

TRY THIS!

Compare a dolphin shape to an ichthyosaur picture. Notice they look similar even though not related!

THINK ABOUT IT:

Why do ichthyosaurs and dolphins look so similar despite not being related?

Chapter 7: The Day Everything Changed

The twins set the time machine to 66 million years ago - the day it all ended.

A bright light streaked across the sky, growing larger and larger.

'The asteroid!' Luna whispered. 'It is about 7 miles wide, moving at 44,000 mph!'

They watched from the safety of the machine as it struck the Earth near Mexico.

The impact was unimaginable - 10 billion times more powerful than a nuclear bomb.

Dust filled the sky, blocking the Sun. Within months, most life on Earth would die.

THE SCIENCE: THE KT EXTINCTION

66 million years ago, an asteroid about 7 miles wide hit what is now Mexico.

The impact created the Chicxulub crater - 110 miles wide and 12 miles deep!

Dust and debris blocked sunlight for months, causing a global 'winter.'

Without sunlight, plants died. Without plants, herbivores starved. Then carnivores.

About 75% of all species went extinct, including all non-bird dinosaurs.

Evidence: a thin layer of rare element iridium found worldwide in 66-million-year-old rock.

SCIENCE FACT BOX

- * The asteroid hit with the force of 10 billion Hiroshima bombs!
- * Tsunamis over 300 feet high swept across the globe after impact.
- * Small mammals survived by burrowing underground and eating insects/seeds.

TRY THIS!

Drop a ball into a tray of flour/cocoa powder. See the 'crater' and 'debris' pattern it makes!

THINK ABOUT IT:

If the asteroid had missed Earth, would dinosaurs still be alive today?

Chapter 8: Back to Today

The twins returned to the present day, their minds full of incredible memories.

'It is amazing that birds are actually living dinosaurs!' Leo said.

'That is right! Birds evolved from small feathered dinosaurs that survived.'

They looked out the window at sparrows hopping on the lawn.

'Those little guys are real-life dinosaurs!' Luna laughed.

'The past teaches us so much about our world,' they agreed. 'Stay curious!'

THE SCIENCE: DINOSAURS LIVE ON!

Birds ARE dinosaurs! They evolved from small, feathered theropod dinosaurs.

The link: archaeopteryx had feathers AND teeth, claws, and a bony tail!

All 10,000+ bird species today are technically avian dinosaurs.

Paleontology is the study of prehistoric life through fossils.

New fossils are found every year, rewriting what we know about the past!

MORAL:

The past teaches us about the future - stay curious and keep exploring!

SCIENCE FACT BOX

- * A chicken is more closely related to T. Rex than a lizard is!
- * Some dinosaur fossils still contain preserved proteins from 80 million years ago.
- * YOU could discover a new dinosaur species - many are still waiting to be found!

DINOSAUR QUIZ

1. What are fossils?

Answer : _____

2. What was Pangaea?

Answer : _____

3. Name the three periods of the Mesozoic Era.

Answer : _____

4. Were pterosaurs dinosaurs? Why or why not?

Answer : _____

5. What was the largest land carnivore?

Answer : _____

6. What killed the dinosaurs 66 million years ago?

Answer : _____

7. What living animals are actually dinosaurs?

Answer : _____

8. How big was Brachiosaurus?

Answer : _____

9. What is a paleontologist?

Answer : _____

10. Name one marine reptile from the dinosaur age.

Answer : _____

DINOSAUR SIZE COMPARISON

Draw a line for each dinosaur's length. Use the scale: 1 inch = 10 feet

Compsognathus: 3 ft (tiny!)

Draw line here: _____

Velociraptor: 6 ft

Draw line here: _____

Human (for comparison): 6 ft

Draw line here: _____

T. Rex: 40 ft

Draw line here: _____

Triceratops: 30 ft

Draw line here: _____

Brachiosaurus: 85 ft

Draw line here: _____

Argentinosaurus: 100 ft

Draw line here: _____

JUNIOR PALEONTOLOGIST CERTIFICATE

I traveled through time and learned about dinosaurs!

My favorite dinosaur: _____

because: _____

Name: _____

Date: _____

Certified Dinosaur Expert!